

Shylendra Bangaru

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Summary

Passionate in leveraging data and technology, a graduate student at Clarkson University skilled in data analysis, visualization, and research. Through roles like Research Assistant, Teaching Assistant, I have showcased adaptability and excellence. Proficient in Python, R, Tableau, and Machine Learning, with expertise in Deep Learning, Data Mining, and virtual reality-based studies. Highly skilled in Salesforce Administration through Trailhead, reflecting the dedication to industry-relevant tools. A quick learner with strong problem-solving abilities and eager to contribute to impactful research and technological innovation.

Education

Clarkson University, MS in Applied Data Science, Jan 2024 - May 2025 - GPA: 3.67/4.0

- **Coursework:** Data Mining, Data Warehousing, Machine Learning, Information Visualization, Natural Language Processing, Deep Learning, Computer Algorithms, Big Data Processing and Cloud Services, Data Driven Application Development, Database Modeling and Design

Amrita Sai Institute of Science and Technology, BTech in Electronics and Communications Engineering, Aug 2018 - Jun 2022 - GPA: 7.77/10

- **Coursework:** Signals and Systems, Electronics Devices and Circuits, Radar Engineering, Optical Communication, Application Oriented Internet of Things, Data Analytics(Elective)

Academic Engagements and Professional Experience

Clarkson University, Potsdam, NY, Jan 2024 – May 2025

- Teaching Assistant – Information Visualization, Probability & Statistics
- Research Assistant – Department of Psychology, Department of Physical Therapy

Neon Enterprises, Hyderabad, Telangana, India, Aug 2022 – Nov 2023

- Worked as a Data Engineer, focusing on designing and optimizing ETL pipelines for efficient data processing and transformation. Developed and maintained scalable database architectures using SQL and NoSQL technologies, ensuring data integrity and accessibility. Implemented data quality checks and monitoring solutions to enhance data accuracy and reporting efficiency. Collaborated with cross-functional teams to analyze business requirements and develop data-driven solutions, leveraging cloud platforms and big data tools to enhance storage, processing, and analytics capabilities.

Technical Skills

- Languages & Visualization Tools: Python, R, Tableau, Matplotlib, HTML5, CSS3, JSON
- Version Control: Git
- Database: MySQL, PostgreSQL, SQLite
- Platforms: Salesforce Trailhead, TensorFlow/Keras, nltk
- Deep Learning: RNNs, LSTMs, GANs

Projects

Early Childhood Development Analysis Using Machine Learning, Clarkson University, Department of Psychology

- Currently leading data-driven research on the Early Head Start Research and Evaluation (EHSRE) Study, applying machine learning and statistical modeling to predict school readiness, academic performance, and behavioral trends in early childhood development. Utilizing Random Forest, XGBoost, and Lasso Regression for predictive analytics, while addressing attrition bias through multiple imputation, inverse probability weighting (IPW), and Full Information Maximum Likelihood (FIML). Optimizing feature selection, model interpretability, and performance via hyperparameter tuning and statistical techniques. This ongoing research contributes to evidence-based policymaking, academic publications, and advanced methodologies in child development analytics.
- Tools Used: R, MATLAB, Tableau, Python, and statistical methods for time-series analysis and data visualization.

Virtual Reality-Based Vestibular Training for Postural Control, Clarkson University, Department of Physical Therapy

- Collaborated on a university-led research project investigating the impact of VR-based vestibular training on human balance and motor control. Conducted data collection using advanced assessments, including video Head Impulse Tests (vHIT), VR simulations, CTSIB, RAMP, and Electrooculography (EOG). Processed experimental data by implementing Python scripts to convert raw data, compute averages and standard deviations, and analyze participants' velocity, acceleration, and consistency across multiple training days. Designed statistical pipelines utilizing ANOVA, t-tests, and correlational analyses to assess motor learning outcomes, highlighting improvements in head velocity control and sensory integration. Managed and analyzed time-series data, including roll-yaw-pitch metrics, to identify performance trends over the study period.
- Tools Used: MATLAB, Tableau, pandas, Google Drive API, Python (for data preprocessing and statistical analysis).

AI-Based Poetry Generator

- Developed an AI-based poetry generator as part of a Natural Language Processing course project, leveraging deep learning techniques to create coherent and creative text. Designed and implemented an RNN-based model utilizing embeddings and fine-tuning techniques to enhance generation quality. Preprocessed large text datasets with tokenization and lemmatization to optimize model training and improve efficiency. Applied LSTM architecture to refine sequential data prediction, ensuring fluency and syntactic accuracy. Focused on building a robust training and evaluation pipeline, demonstrating a practical application of deep learning in natural language generation.
- Tools Used: TensorFlow/Keras, Transformers, NLTK, pandas, NumPy, and Matplotlib.

Improved GAN for Image Generation and Classification

- Enhanced the training stability of Generative Adversarial Networks (GANs) using techniques like feature matching, minibatch discrimination, and virtual batch normalization.
- Applied the model to large datasets such as CIFAR-10, SVHN, ImageNet, and MNIST, achieving high-quality image generation and improved semi-supervised classification performance.
- Evaluated model outputs using metrics like the Inception Score and visual Turing tests, showcasing advancements in realism and diversity of generated images.
- Tools Used: TensorFlow, Python, NumPy, and Matplotlib.

Obesity Level Prediction Using Data Mining Techniques

- Developed a data mining model to predict obesity levels using a dataset of 2111 records and 17 attributes, including Age, Gender, BMI, Physical Activity Frequency, Daily Water Intake, Family History of Obesity, and Eating Habits. Conducted statistical analysis, such as Chi-squared tests and ANOVA, to identify significant factors influencing obesity risk. Built predictive models using Multiple Linear Regression (MLR) and analyzed feature correlations to enhance model accuracy. Applied data preprocessing techniques to convert non-numeric variables into numerical formats, ensuring effective model training and improving prediction reliability.
- Tools Used: Python (pandas, NumPy, scikit-learn), Matplotlib, and statistical modeling techniques.